

Roll No

AT-703(C)-CBGS

B.Tech., VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Instrumentation

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) What is BJT? Explain its operating regions.
b) What is semiconductor? Describe p-n junction.
2. Define Rectifier. Explain Half wave rectifier and Full wave rectifier.
3. a) What is Op-amp? Write down ideal op-amp characteristics.
b) What is h-parameter model? What is the h-parameter model of transistor?
4. For the following logic equation: $Y = KL' + L'M + KM$
 - i) Draw the complete logic diagram directly from the logic equation.
 - ii) Give the complete truth table.
 - iii) Give the canonical form SOP and POS.
 - iv) Give the Minterm and Maxterm representations.

5. a) Assume $V_{REF}=10V$ and $R=10K$ ohm. Determine the resolution and full-scale output for this DAC. Assume that R_L is much smaller than R .
b) Explain the operating principle of this analog to digital converter circuit, usually referred to as a successive-approximation converter.
6. a) What is LVDT? Explain in detail.
b) How do strain gauges work? What are the practical purpose and applications of strain gauge?
7. a) Differentiate between the following:
 - i) AC bridge and DC bridge
 - ii) Maxwell bridge and Hay's bridge
b) What is a Tacho-generator and its types? How does it work? Explain.
8. Define the following:
 - a) Clipper and clamper
 - b) K-Map
 - c) Potentiometer
 - d) Boolean Algebra
